

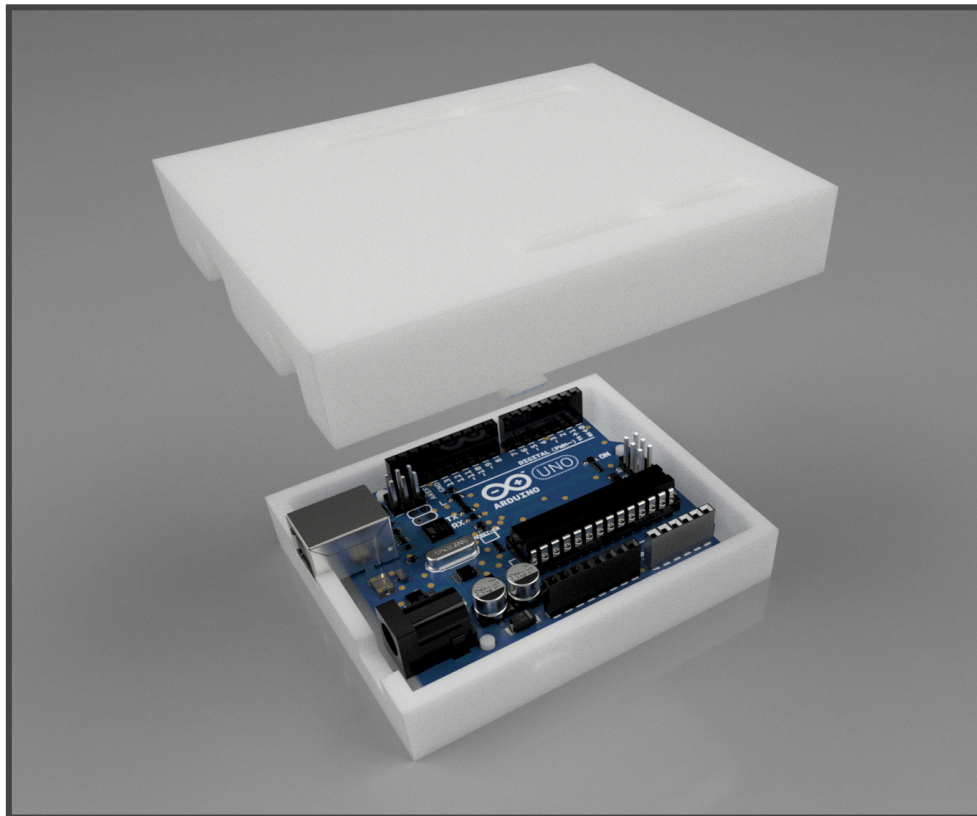
3D Printing and Modeling

11th lab: Modeling around electronic components

0. Resources

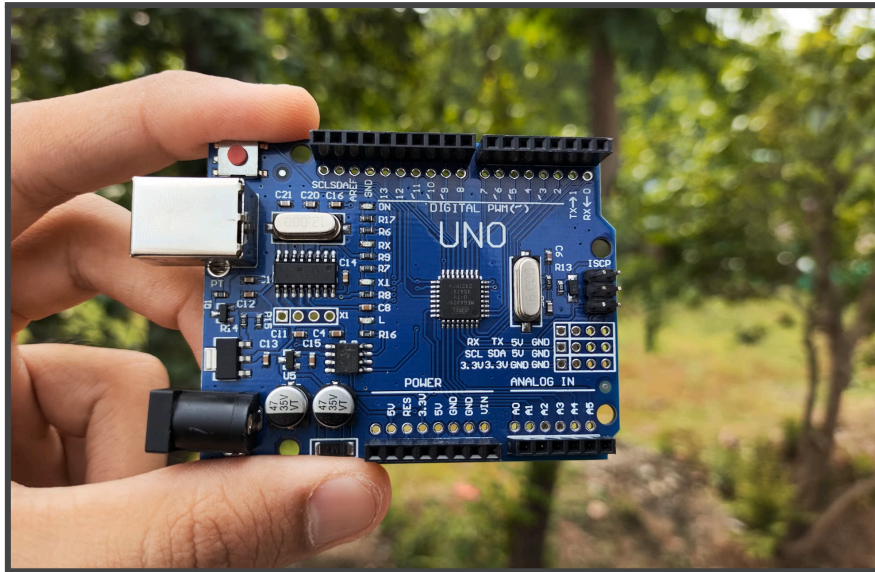
1. [Designing a 3D Printed Enclosure for Arduino Uno](#) (Similar Project)
2. [How to Design a Print with Perfect Tolerance](#) (13 min)

1. The Arduino Case



The purpose of this laboratory is designing a case for an Arduino Uno board. We aim to showcase the principles behind designing around electronic components so you can freely include these in your projects.

1.1. The Arduino Board

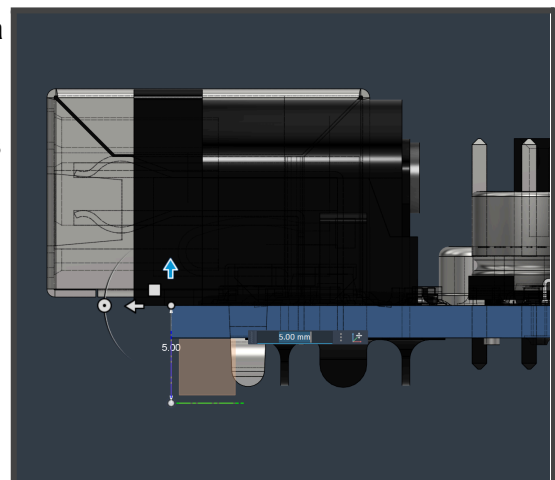


Arduino is an open-source electronics platform made of easy-to-use hardware and software that lets you build interactive projects by connecting sensors, lights, motors, and other components to a programmable board. It's popular for beginners, hobbyists, and makers because it makes it simple to turn code into real-world actions. Starting from a similar board, we aim to create a case around it in fusion.

Fusion 360 files for Arduino boards are freely available online. This is very helpful as it helps us save time when designing for it. If that wasn't the case, our task would begin by measuring the components we are designing for and recreating them in Fusion. Today all you need to do is to open the .f3d file from Teams and import it into your design.

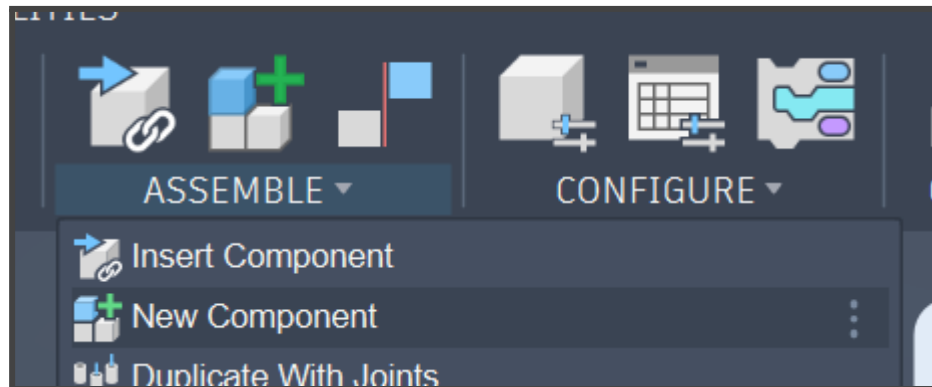
We will start by creating a new hybrid design file (assembly can work, but hybrid makes it easier to switch between components) and importing (**Insert > Component**) the component with the Arduino board. You can play around with it to see what makes up the board.

You will then need to move the board into a more comfortable position, such that **Top** and **Bottom** are properly aligned. You should also keep in mind that the board has many solder joints on the bottom, which also need to fit inside the case. **Move the board 5-6 mm above the XY plane.**

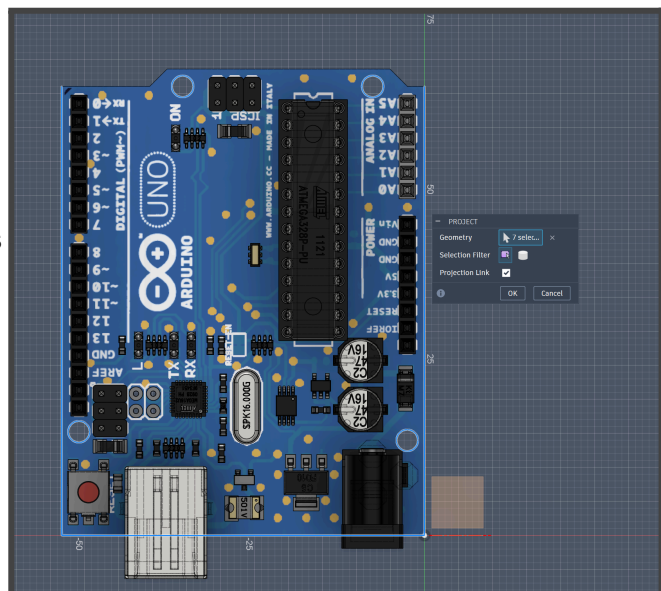


1.2. The case

We can now start working on the case. We'll start by creating a new component to house all of our sketches and bodies. Inside this new component, create a sketch on the XY plane to start designing our case. (**Make sure to activate the desired component**)

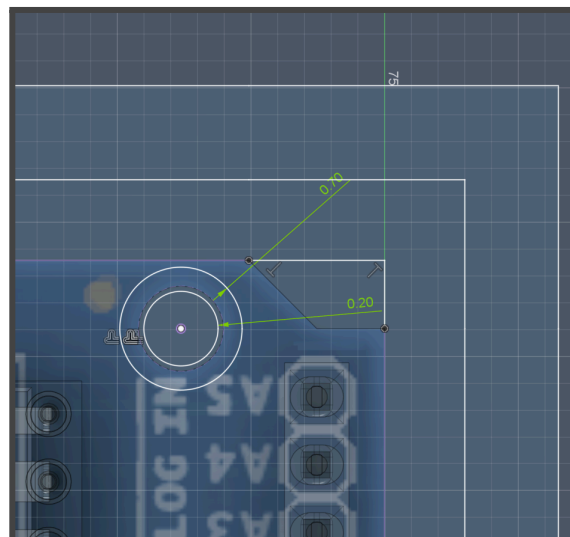


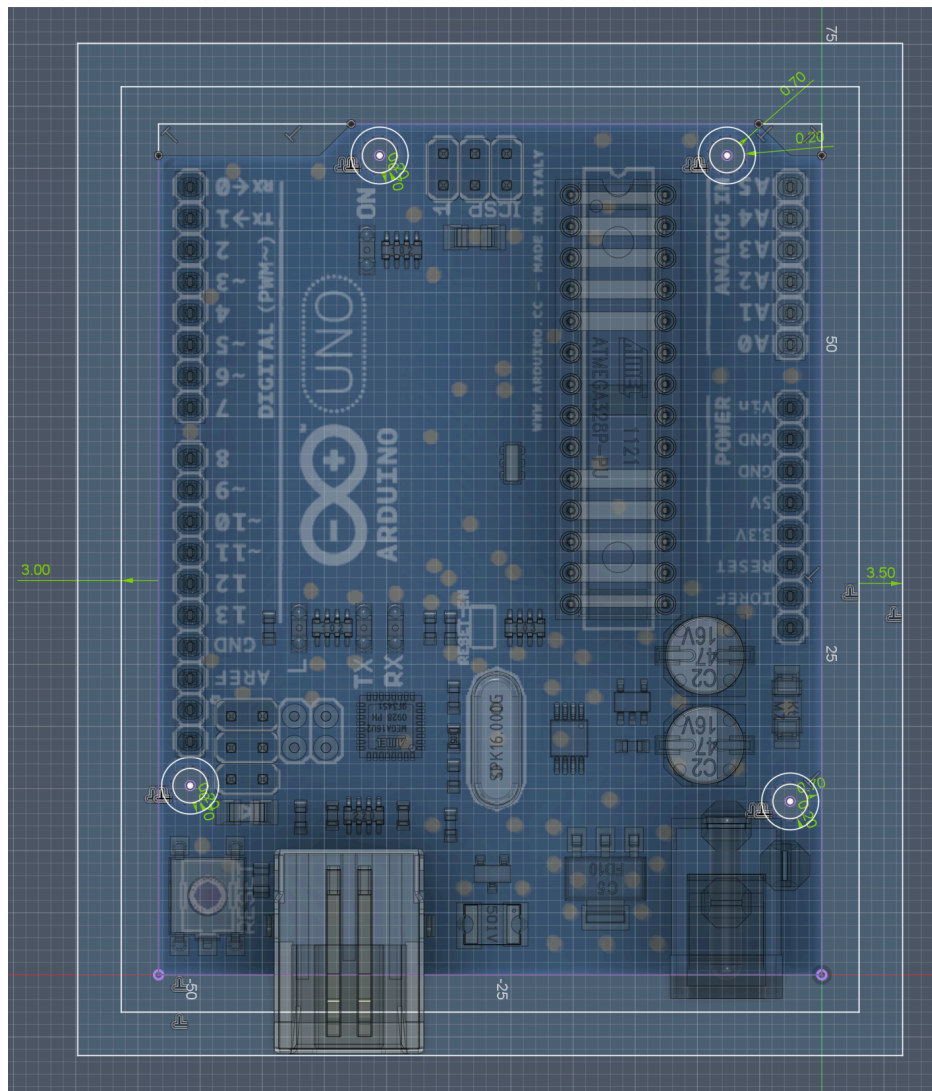
Your sketch should look similar to the image on the right. The point of importing the board into Fusion 360 was to use its existing dimensions. In order to do that we will take advantage of the **Project** tool (**Create > Project**) to help us bring geometry from the board into our sketch. We are interested in the outside shape of the board, and the four circular mounting holes. Click **OK** after selecting them.



You can then adjust the geometry to make a more box-like, rectangular shape. The **offset** tool is your best friend here. You can add a 3mm offset around the perimeter to make space for the board to fit inside the case, then another 3.5mm offset to act as walls.

Finally, we will use the mounting holes to keep the board in place. We will again need two offsets, one smaller than the diameter of the projection and another one slightly bigger, acting as a stopper for the board.

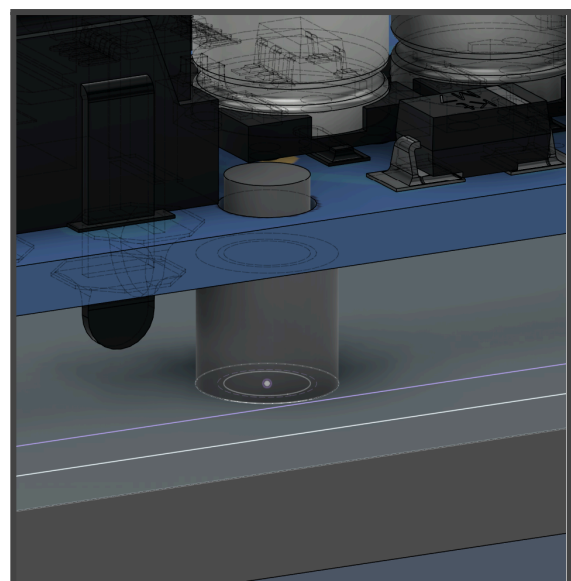


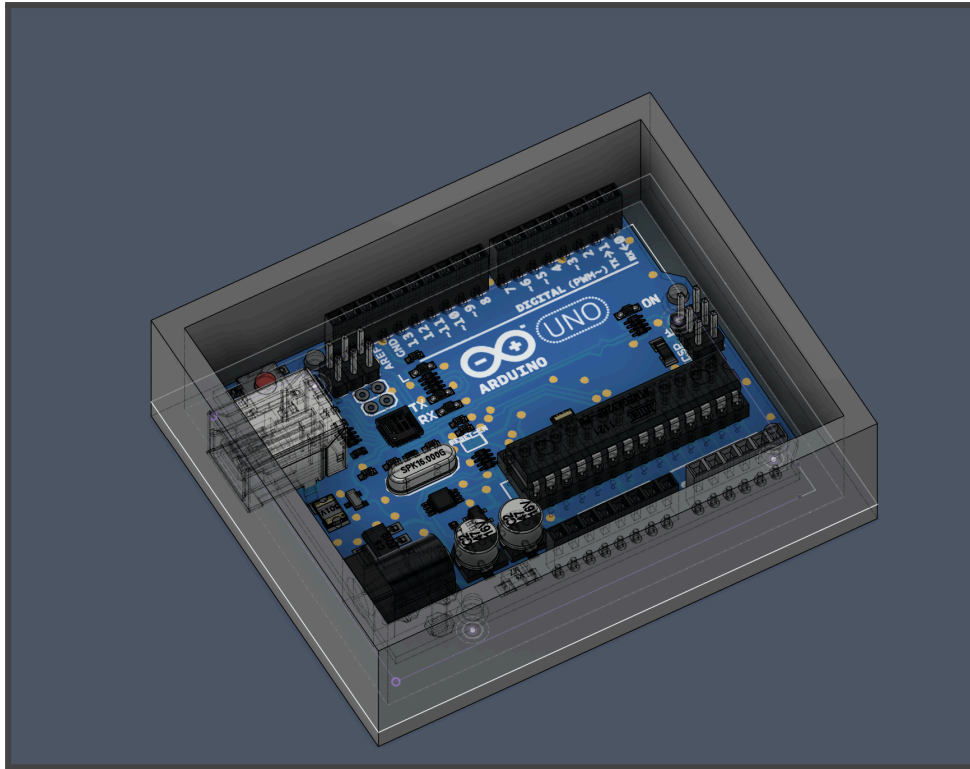


When finished, the sketch should look something like this.

We can now use this sketch to extrude the bottom part of our case, including the mounting pins. They should look similar to the image on the right, where the outer diameter circle goes up to the height of the board, and only the inner circle goes through the board.

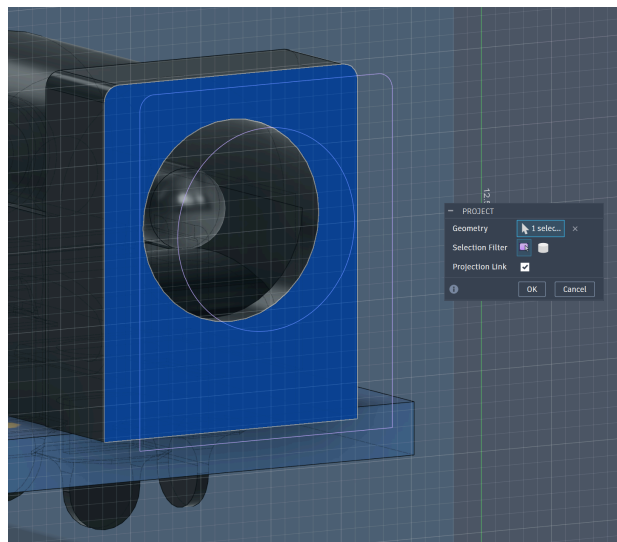
If you struggle understanding which component goes where, you can always toggle visibility or even set the opacity of each component in the browser.



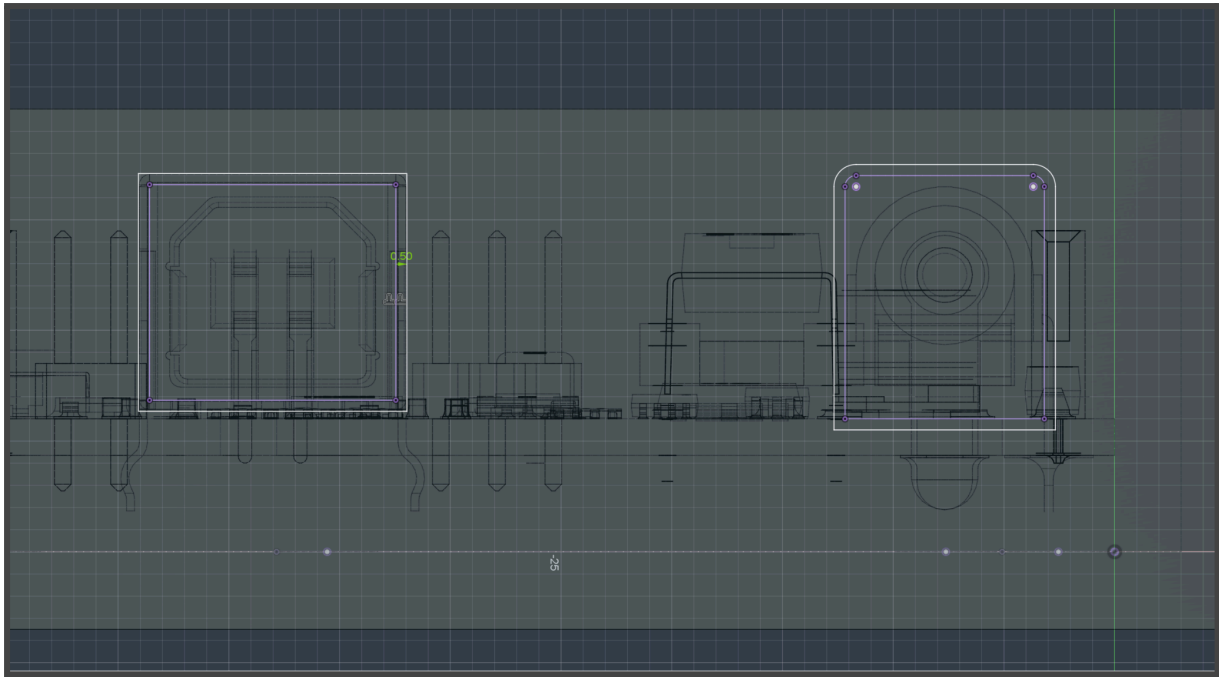


Once you are finished extruding the case, you can probably see there is a slight problem with our design. There is no way to access the USB port or the power jack. The next step would be to make the appropriate cutouts for those two.

Create a **new sketch** on that specific side of the board. Once again, we will be using **Project** to get a rough shape of the ports.



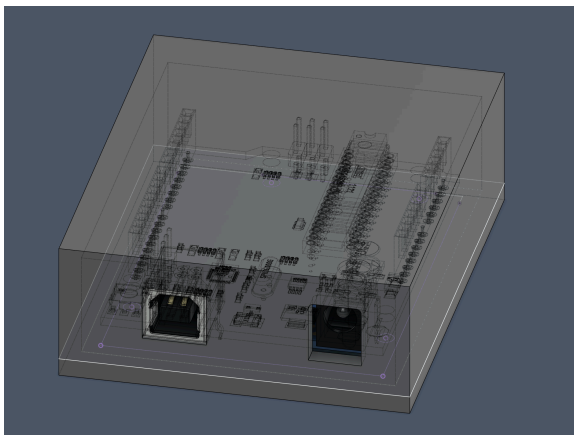
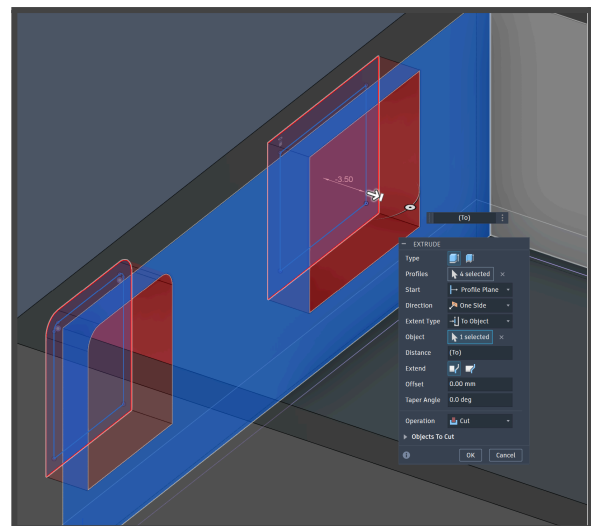
You can either select each edge, or an entire surface you want to project. Do the same for both ports. After projecting, use offset to enlarge the shape so it fits well every time.



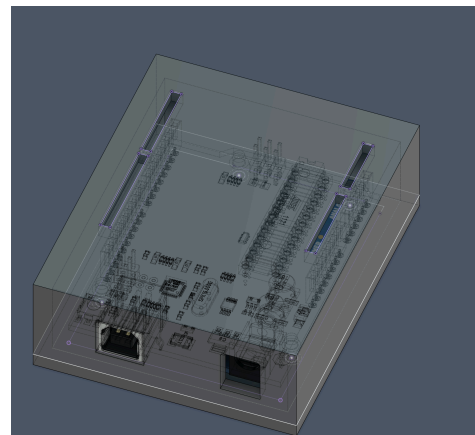
Your result should look like this

Now make the cuts. You can extrude to the dimension of your walls, or use **Extrude To Object** and select the inner wall.

Next, we will repeat the process for the top of the case. Create the top if you didn't already, and make the cuts for the **GPIO pins**.

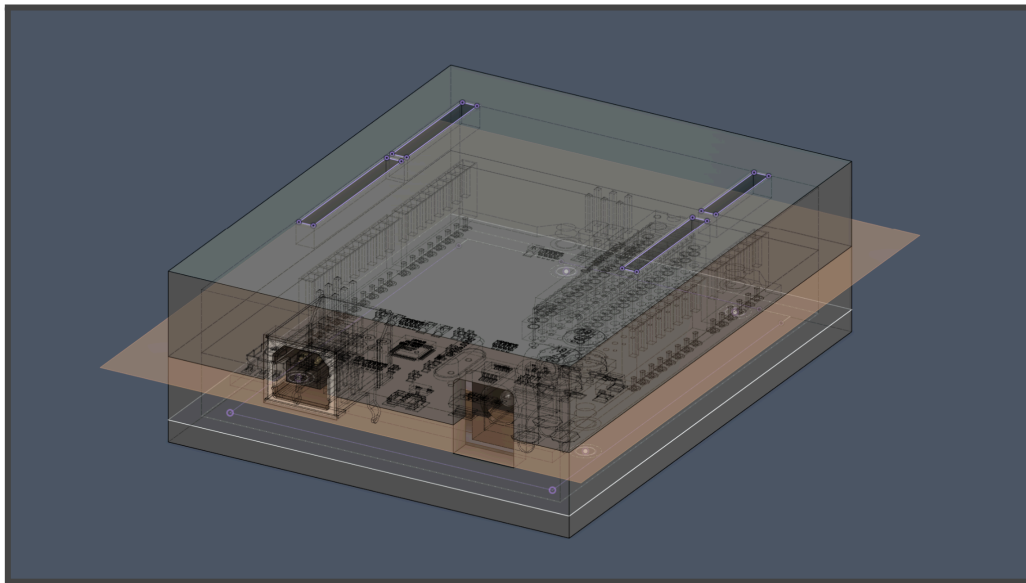


Before the cuts



After the cuts

1.3. Splitting the case

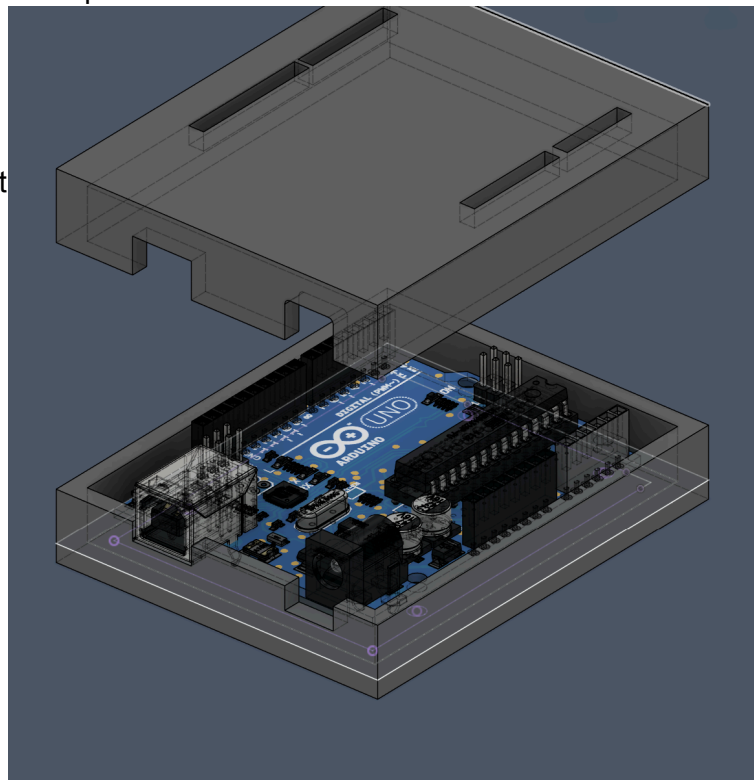


Right now, our case is only composed of one body, which can not be taken apart and reassembled. This is a great occasion to learn about a new tool, called **Split Body** which allows us to cut a body into two, by using a given geometry as a cutting tool. For our purposes, a simple cut is just fine.

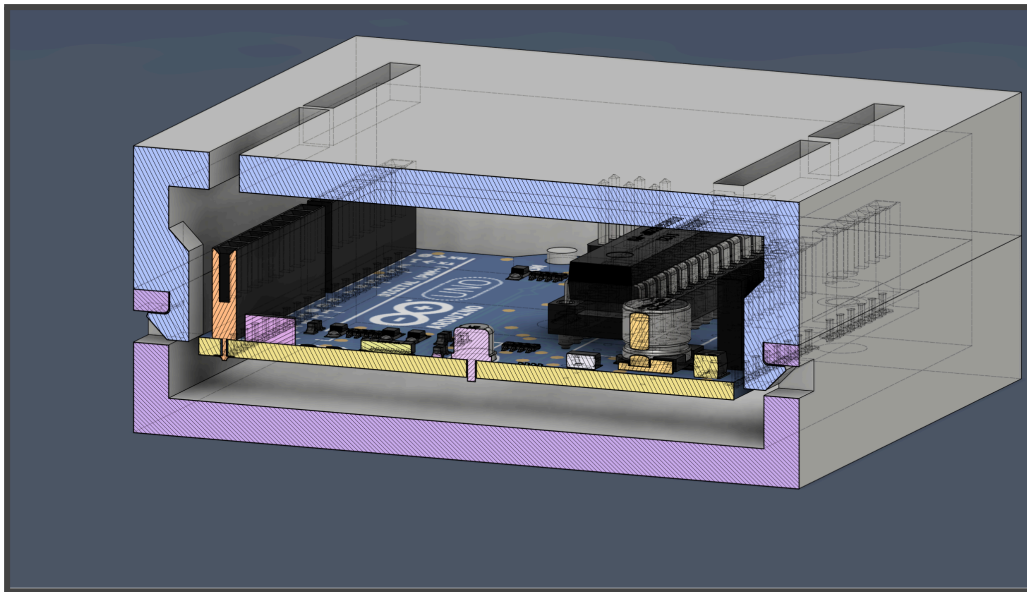
Start by creating a **Midplane** that cuts horizontally through the component, as shown in the image above. This plane will act as the knife when splitting the part. Go to **Modify > Split Body**, select the body and the new plane as **tool** and click OK. You should now see two bodies in the browser instead of one.

Finally, you can now turn each of your bodies into components if you wish. This is not mandatory for what we will be doing next.

Now, the case is a lot more functional. The only problem left is that there is no way to attach the top and bottom cases.



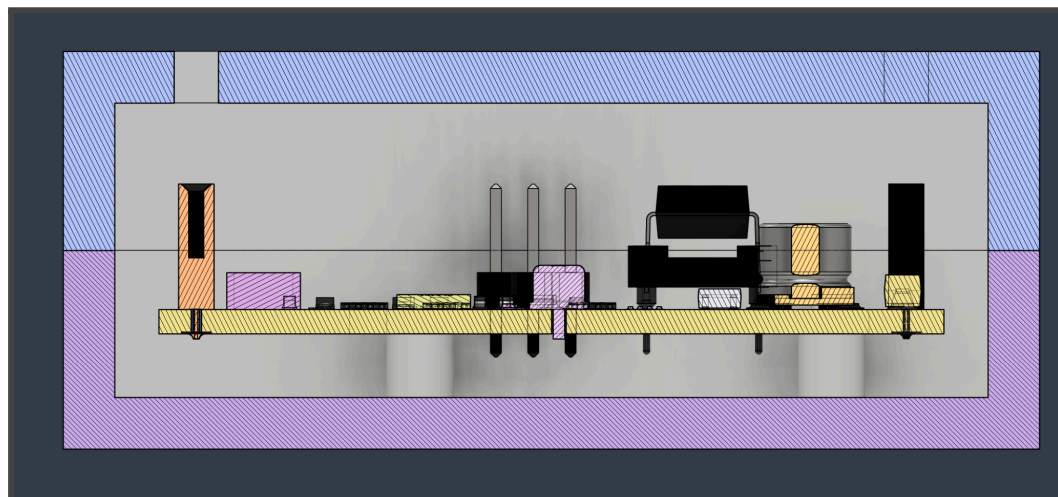
1.4. Snap Joints



We will now attempt to recreate a snap fit mechanism for our case. This type of design allows for the two parts to snap into place, but also be taken apart when needed.

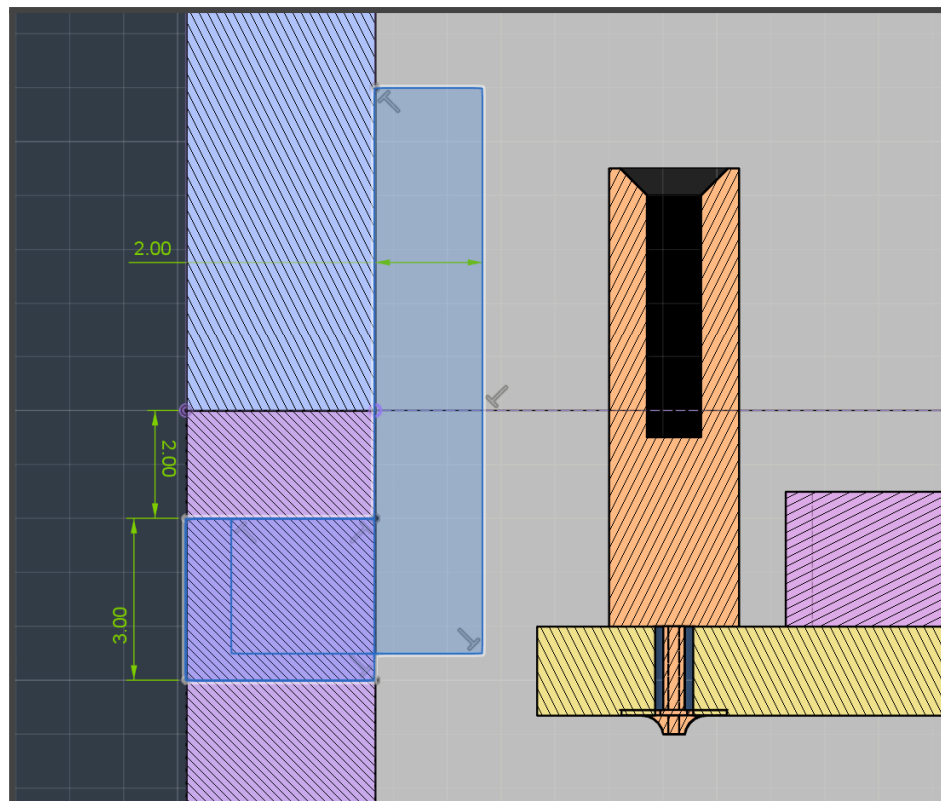
To do that, we will start by getting a better look inside the case, to see how much space we have left and where. Go to **Inspect > Section Analysis**, then **select a plane parallel to how you want to section your objects**, to create an analysis that allows us to “cut” into the case with our vision. At any time, you can edit the position of this analysis at any time by right clicking it in the browser, or disable it.

Use the analysis to find a section across the case that might have enough space to accommodate the clips.



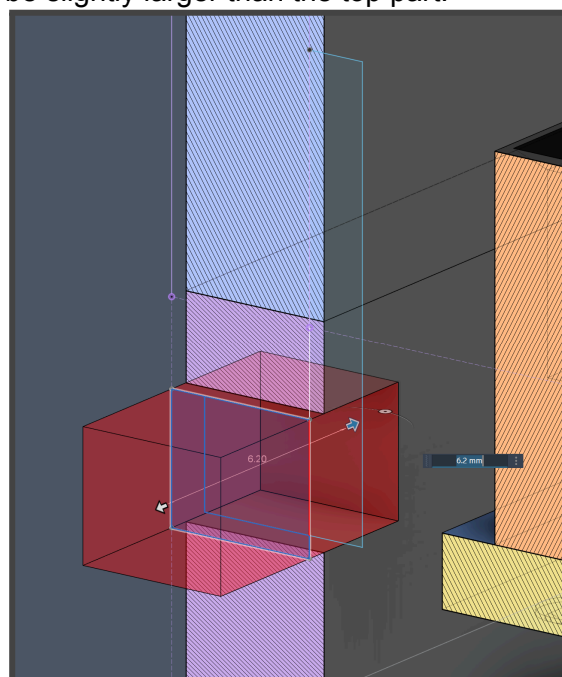
It would probably look something like this. When ready, create a new sketch in that position.

You might need to create a new **Offset Plane** and a new sketch onto it, as you might not be able to find a surface on which to place your sketch.

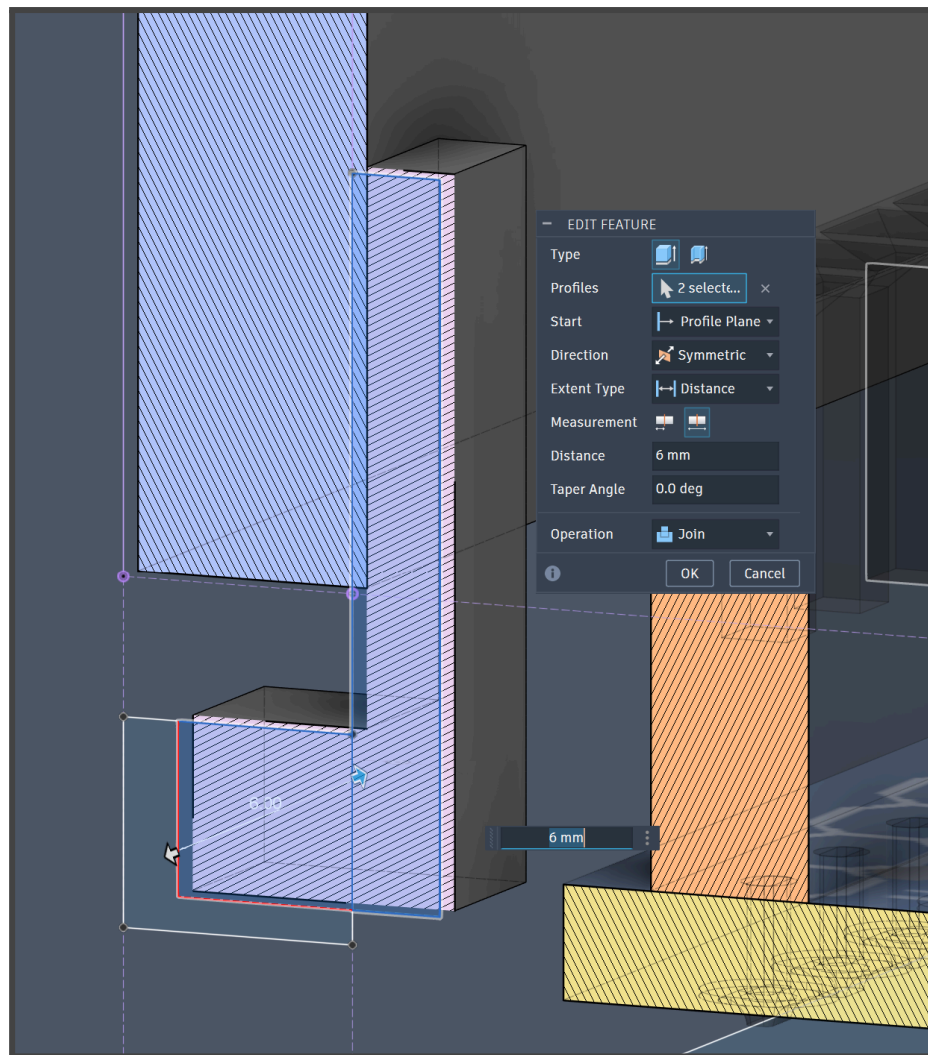


In this sketch, draw a shape similar to this. The inverse L shape will be part of the top case and is supposed to bend and snap into place, holding the bottom case in position.

The larger, 3mm, rectangle is the shape of the slot we need to cut into the bottom case. This slot needs to be slightly larger than the top part.



Extrude the two shapes accordingly. Start by cutting into the bottom section. This extrude will also be made .2 mm longer than the length of the snap joint (6.2 vs 6 mm) for a better fit.

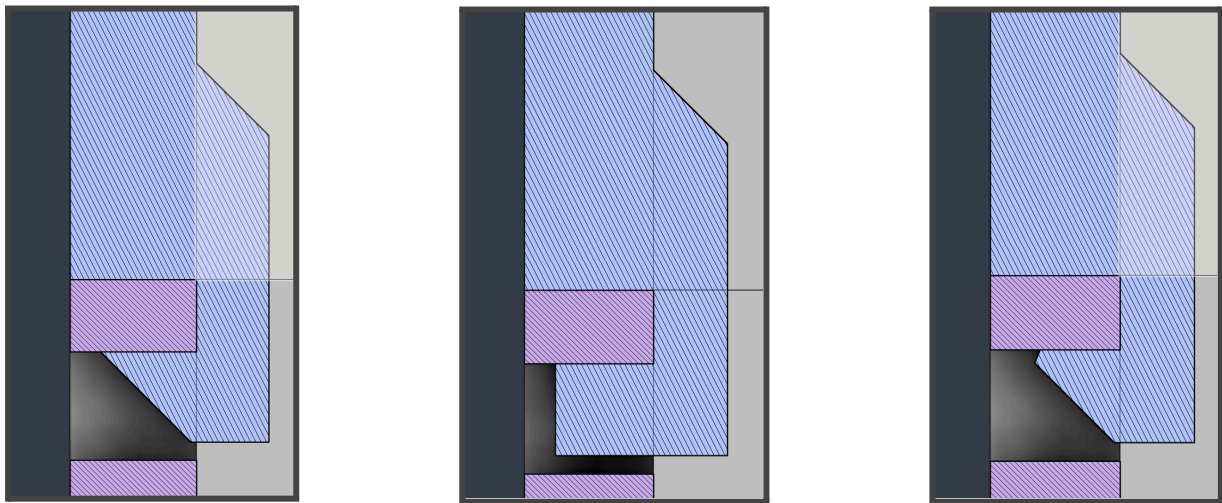


You can then extrude the top part with operation type Join.

When extruding **Extrude** with operation **Join** or **Cut**, Fusion might not always understand which body you want to join or cut into. The easiest way to fix this is to hide the unwanted bodies from the browser. In the image above I only kept the top case visible.

If you still cannot join the two bodies when extruding, you can unite the afterwards using **Modify > Combine**.

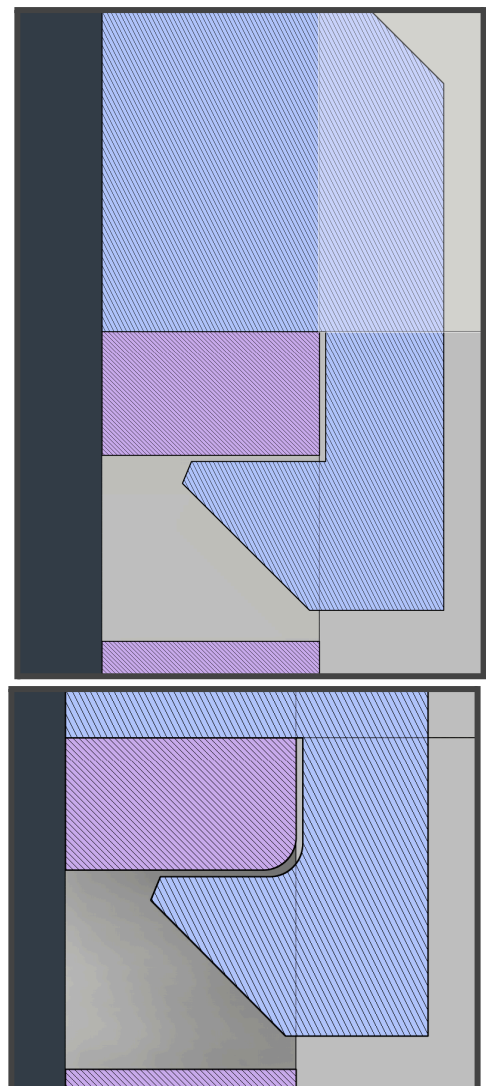
1.5. Finishing touches

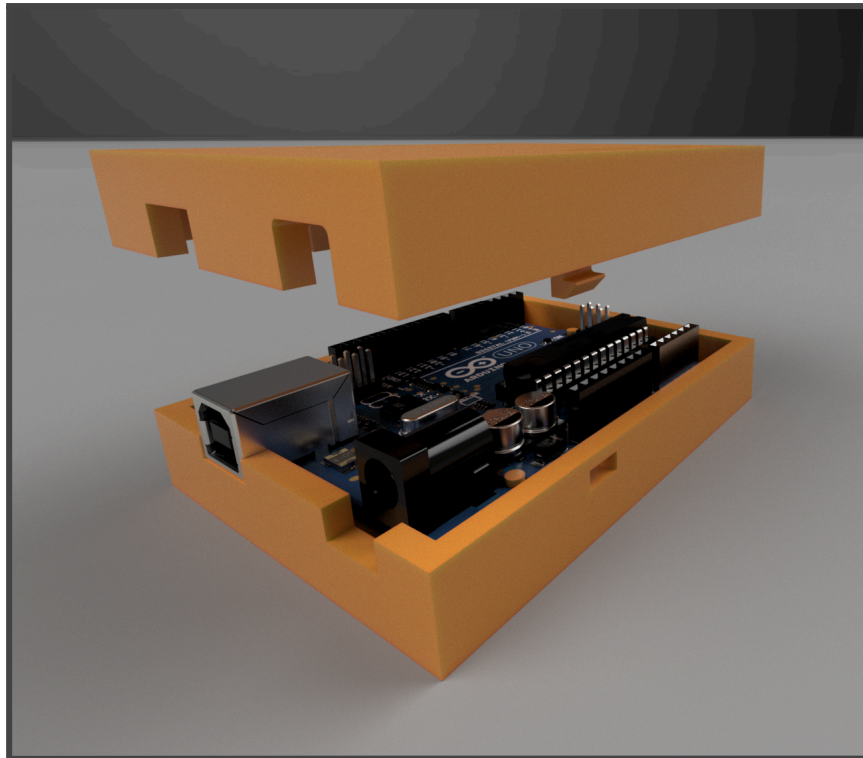


Only a few more structural chamfers are needed. Can you guess what the purpose of each one is?

Next, offset the two faces that are touching the bottom case. .1 mm should be enough. These will provide some printing tolerances.

Lastly, fillet the inner corners. As discussed earlier, filleted corners are relatively stronger when 3D printed compared to 90 degree ones.





Congratulations! You have now finished the case. You can still add many things such as rounded corners on the outside to make it both easier to print and nicer in hand.